SCHOOL OF ELECTRONICS ENGINEERING

Cordially invites you to attend the Ph.D. Colloquium

on

Deep Learning Architectures for Automated

Analysis and Classification of ECG Waveforms

By

PREMANAND S

Register No: **19PHD1047** Category: **IFT**

Date: **18-03-2026** Venue: **AB1- 806** Time: **3.00pm**

Abstract

Electrocardiogram (ECG) signal analysis plays an important role in diagnosing and monitoring cardiovascular conditions. With the rapid growth of biomedical signal data, automated ECG classification using artificial intelligence has become increasingly important for supporting clinical decision-making. However, challenges remain in effective feature extraction, computational efficiency, and capturing both local and long-range dependencies in ECG signals. This research focuses on developing hybrid deep learning frameworks that enhance feature representation and improve ECG signal classification performance. Initially, a feature extraction framework termed Convolution-based Heterogeneous Activation Facility (CHAF) is proposed, which employs heterogeneous activation functions across convolutional blocks to capture diverse signal characteristics and improve feature learning. To improve capturing the global morphological patterns, an advanced hybrid architecture called ECG-ResViT integrates Convolutional Neural Networks (CNNs) with Vision Transformers (ViTs) to capture both local morphological patterns and long-range dependencies in ECG signals. Further enhancement is achieved through BlendNet, which combines complementary ECG representations using an alpha-blending mechanism to generate richer feature maps for improved classification. Collectively, this research proposes hybrid deep learning architectures that improve feature representation, classification accuracy, and computational efficiency for automated ECG signal analysis.

Publications:

1. **S. Premanand and S. Narayanan**, "BlendNet: A blending-based convolutional neural network for effective deep learning of electrocardiogram signals," *Frontiers in Artificial Intelligence*, vol. 8, p. 1625637, Aug. 2025. (Scopus, Q1, JCR 4.7)
2. **S. Premanand and S. Narayanan**, "ECG-ResViT: A hybrid CNN-ViT model for efficient ECG signal classification," *Eur. Phys. J. Spec. Top.*, vol. 234, no. 15, pp. 4483–4496, Oct. 2025. (Scopus, Q2, JCR 2.3)
3. **S. Premanand and S. Narayanan**, "Convolution-based heterogeneous activation facility for effective machine learning of ECG signals," *Comput. Mater. Contin.*, vol. 77, no. 1, p. 25, 2023. (Scopus, Q2, JCR 3.1)

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